



Incompatibilities in formulation



Outline

- Introduction
- Importance of determining incompatibility
- Types of incompatibility:
 - Chemical incompatibility
 - Pharmaceutical or physical incompatibility
 - Therapeutic incompatibility
- Methods of rectifying incompatibility

Objectives

- After the end of this topic ,students will be able to :
 - Know the importance of determining incompatibility
 - Describe the different types of incompatibility
 - Discuss the different methods used for rectifying incompatibility

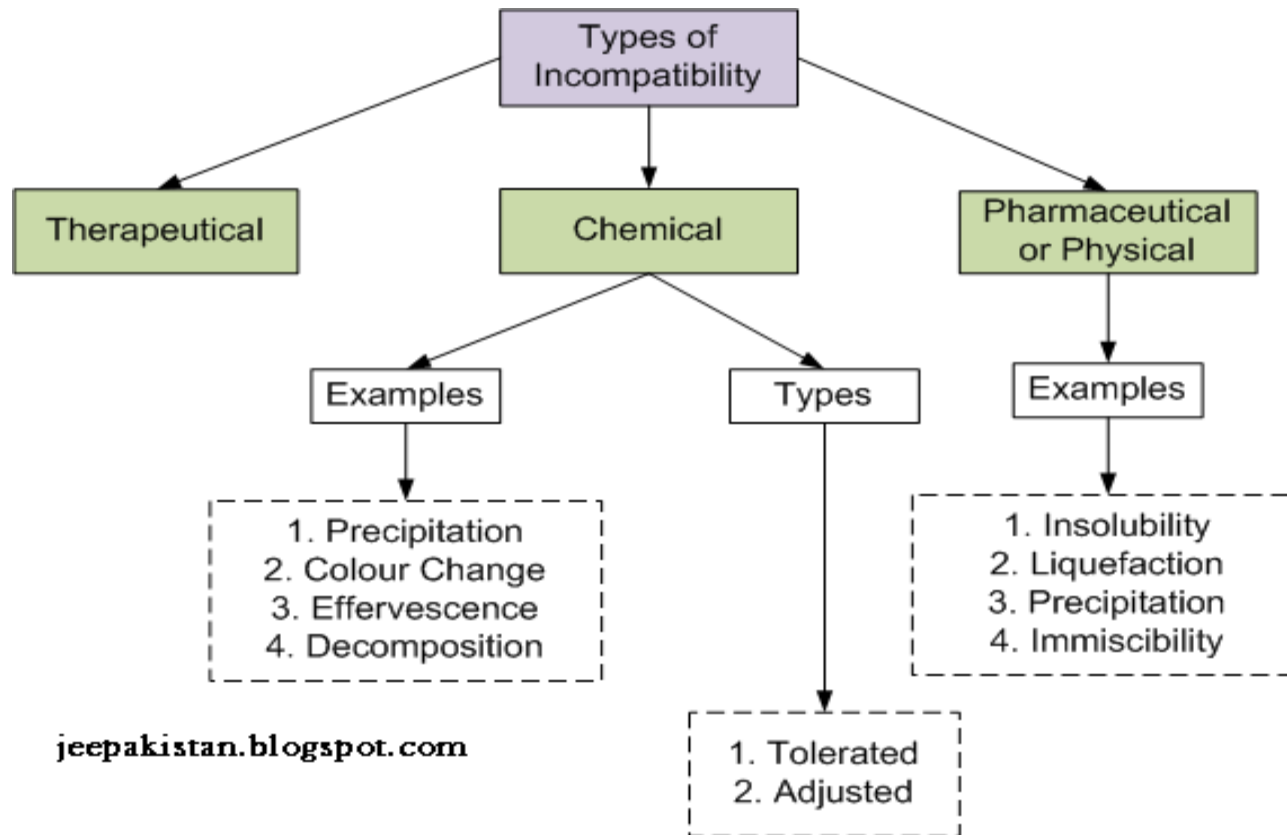
Introduction

- Incompatibility refers to the inability of something or some process to co-exist with another process or thing.
- Pharmaceutical incompatibility refers to the inability of a pharmaceutical substance to exist in combination with another pharmaceutical entity.
- The term incompatibility may be applied to the pharmaceuticals when problem arises during their compounding, dispensing or administration.
- Incompatibilities were frequent in earlier days when the prescription contains multiple ingredients.
- There are three types of incompatibility:
 1. Therapeutic incompatibility
 2. Chemical incompatibility
 3. Pharmaceutical or physical incompatibility

Importance of Determining Incompatibility

- Incompatible products may affect:
 - Ø Safety of medicament
 - Ø Efficacy of product
 - Ø Appearance of a medicine
 - Ø Purpose of medication

- The most important step in dealing with incompatibility is recognizing by noting an incompatibility before a prescription is filled.
- The pharmacist can take corrective measures that save both time and material.
- It is easier to prevent an incompatibility rather than to correct it.
- An incompatibility is not recognized until after the prescription has been compounded.
- The prescription should not be dispensed until the incompatibility has been corrected.



jeepakistan.blogspot.com

Chemical incompatibilities

- **Chemical incompatibilities** often occur due to oxidation-reduction, hydrolysis or combination reactions.
- A chemical incompatibility, which is visualized rapidly by effervescence, precipitation or color change, is called **immediate incompatibility**.
- An incompatibility without immediate and visible physical change is known as **delayed incompatibility** and may or may not result in loss of therapeutic activity.

- Unintentional incompatibility may be:
 - tolerated i.e. interaction is minimized but composition of the prescription is not altered;
 - adjusted i.e. interaction is prevented by addition or substitution of constituents without affecting the medicinal action of the preparation.
- It is recommended that the pharmacist must consult or notify the prescriber about the suggested modification in the prescription while overcoming the incompatibility.

Chemical incompatibility:

This type of incompatibility is resulted due to the formation of undesirable new product when two or more drugs are combined.

Examples of Chemical incompatibility:

- Hydrolysis
- Oxidation
- Polymerization
- Isomerization
- Decarboxylation
- Absorption of CO₂
- Combination
- Formation of insoluble complexes

Types of Chemical Incompatibility:

Chemical incompatibility can be intentional i.e. a prescriber knowingly gives incompatible drugs, or unintentional i.e. prescriber does not know that the drugs are incompatible.

Chemical incompatibilities may be classified as follows:

Oxidation:

- Oxidation refers to the loss of electrons and reduction to the gain of electrons.
- Prescription mixtures are usually oxidized on exposure to air, higher storage temperatures, light, over dilution, incorrect pH adjustment or in presence of catalysis.
- Use of **antioxidants** such as ascorbic acid, sodium sulphite or sodium metabisulphite is often helpful.

- Trace metal ion catalysis may be counteracted by complexing agents such as **disodium edetate** and sodium calcium edetate.
- Auto-oxidation in fats and oils, phenolic substances, aldehydes and vitamins is controlled by agents such as propyl gallate, thymol, butylated hydroxyanisole (BHA), butylated hydroxytoluene (BHT), and hydroquinone etc.
- Silver, mercury and gold salts may be reduced by light to the metallic form although such reactions are rare in prescriptions.

Oxidation-loss of electrons or gain of oxygen

Factors leading to oxidation:

- Presence of oxygen
- Light- triggers photo-chemical reactions
- Temperature- elevated temperature accelerate oxidation reaction
- pH
- Type of dosage form- oxidation reaction occur in solutions faster than in solid

Factors leading to oxidation...

- Presence of pre-oxidants as metals and peroxides
- Type of solvent used- oxidation reaction occur faster in aqueous solution than others.
- Presence of unsaturated bonds- double and triple bonds in unsaturated fatty acids (oils) undergo oxidation easily than the oils containing saturated bonds (margarine)

Protection of drugs from oxidation

- Addition of Antioxidants: Vitamin E, vitamin C and inorganic sulfur compounds such as thiosulfate and polysulfide
- Addition of chemicals which form complexes with metals i.e. EDTA, Benzalkonium chloride
- Protection from light by using a dark container and storing drug formulations in dark places
- Packaging with substances which absorbed light i.e. Oxybenzene

- Choice of suitable pharmaceutical dosage forms which reduce the possibility of oxidation, like solid dosage forms versus solutions
- Maintenance of pH using buffered solutions
- Choice of suitable solvent (rather than water)
- Storage at low temperature
- Protection from air by:
 - a. using good closed containers
 - b. Replacement of oxygen by nitrogen

Chemical groups subjected to oxidation

- Phenolic compounds: Phenylephrine
- Catechol derivatives: Adrenaline and noradrenaline
- Some antibiotics: Tetracyclines
- Oils (fixed and volatile)
- Vitamins (lipid and water soluble)

How to detect oxidation

- Change of color, odor, viscosity of the formulation.
- For fixed and volatile oils: rancidity change of color, taste, odor, and viscosity.

B. Acid-base reactions:

Such reactions result into precipitation, gas formation, colour development or colour change.

Change in or Development of Colour:

Most of the dyes employed in pharmaceutical practice and their colour are influenced by their ionization depending on pH of the solution.

Laxative phenolphthalein is colourless in acid solution but red in alkaline mixtures.

- Gentian violet is a basic purple compound but on addition of acid, the compound changes the colour through green to yellow.
- Such incompatibilities are corrected by the addition of a buffer or change of the vehicle to prevent formation of free acid or base from the salt.

C. Hydrolysis:

Many substances hydrolyze in water and their reaction may be facilitated by heat catalysts, hydrogen ions and hydroxyl ions.

Esters, amides and metals like Zn and Fe etc., are common examples.

Soluble salts of barbituric acid derivatives and sulphonamides hydrolyze in water and yield insoluble free acids.

Phenyl salicylate hydrolyses in basic media to salicylic acid and phenol.

- Addition of any of the species formed as a result of hydrolysis is a common method employed to prevent or reverse the ionic hydrolysis.
- Examples of drug substances which may undergo hydrolytic decomposition include procaine, sulphonamides, chlorothiazide, barbituric acid derivatives, aspirin, some alkaloids, and penicillin.
- Similarly gelatin, sucrose, sodium acetate, flavouring oils and chlorobutanol; some of the common ingredients in prescriptions; are also liable to decomposition by hydrolysis.

E. Racemization:

- It is the conversion of an optically active form of a drug substance to an optically inactive form without a change in chemical constitution but is usually associated with a reduction in pharmacological activity.
- Examples of substances undergoing racemization are **adrenaline, ephedrine, norephedrine**.
- In alkaline solution but not in acid solutions, 1-hyoscyamine may undergo racemization to form atropine.

Physical Incompatibility

Physical incompatibilities are often called:

pharmaceutical incompatibilities.

- This type of incompatibility results by the slow or immediate formation of decomposed solutions or precipitates, when the drugs are combined in a pharmacy setup or laboratory.
- *Def.:* Interaction between two or more substances which lead to change in color, odor, taste, viscosity and morphology.

Manifestations of physical incompatibility:

The following list outlines the various ways incompatibility between or among drug agents may be manifested.

1. Insolubility of prescribed agent in vehicle (When 2 or more substances are combined they do not give homogenous product due to immiscibility or insolubility.
2. Immiscibility of two or more liquids
3. Liquification of solids mixed in a dry state (called eutexia)

1. Insolubility:

The following factors affect the solubility of prescribed agent in vehicle and may render it less soluble:

1. Change in pH
2. Milling
3. Surfactant
4. Chemical reaction
5. Complex formation
6. Co-solvent

Any change in previous factors may lead to precipitation of drugs and change in their properties.

Example 1:

Rx

Benzalkonium chloride

Sodium lauryl sulfate

They are not mixed together because benzalkonium chloride is positive charged while sodium lauryl sulfate has negative charge.

By mixing together a precipitate is formed.

Example 2:

Rx

Ephedrine sulfate

Menthol

Liquid paraffin

This prescription is not prescribed because ephedrine sulfate is a salt which is soluble in water but insoluble in organic solvents, oil and paraffin.

Example of prescription:

Terpine hydrate 3g

Simple syrup q.s to make 120ml

Terpine hydrate is insoluble in simple syrup then how we make solution? So in this case $\frac{1}{2}$ of the syrup may be replace by alcohol or iso-alcohlic elixirs or the drug may be suspended with accasia tragacanth and dispense with a shake label.

Example of prescription:

Magnesium carbonate 3.75g

Citric acid 7.5g

Sodium bicarbonate 7.5g

Boiled water q.s to make 250ml

- A clean and complete solution is not produced, since some of the magnesium carbonate will remain unchanged.
- A perfectly clean solution can be obtained by adding the magnesium carbonate to the solution of citric acid first and allow these 2 compounds to react completely to form solution of magnesium citrate and then adding the sodium bicarbonate.
- The reason for this order of mixing is that both magnesium carbonate and sodium bicarbonate will react with citric acid in solution. When bicarbonate ion is added first it uses some of citric acid so that not enough is left to convert all the magnesium carbonate to solubilize magnesium citrate. As a result the carbonate that is left will not dissolve by reacting the magnesium carbonate and citric acid.
- First all of the carbonate is converted to the citrate then the sodium bicarbonate which is water soluble is added to give a clean solution.

Insolubility

This can be corrected by using one or more of the following methods:

Addition of suspending agents or thickening agents:

- In the following prescription tragacanth (mucilage or compound powder) is used as a suspending agent.

Phenacetin	3g
Caffeine	1g
Orange Syrup	12ml
Water up to	90ml

- As Phenacetin is an indiffusible substance.

- 1) Some insoluble powders such as sulphur and certain corticosteroids (*hydrocortisone acetate*) and antibiotics are difficult to wet with water.

Remedy: Wetting agents

e.g. *saponins* for sulphur containing lotions

and *polysorbates* in parenteral suspensions of corticosteroids and antibiotics are used to distribute the powder

- 2) When a resinous tincture is added to water the water insoluble resin agglomerate forming indiffusible clots.

Remedy: This is prevented by slowly adding the undiluted dispersion of protective colloid (*Tragacanth mucilage*).

e.g. Lobelia & Stramonium tincture which should be mixed with tragacanth mucilage and stirred constantly. This will produce a stable preparation.

3) Liquid preparations containing indiffusible solids such as chalk, aromatic chalk powder, succinyl sulfathiazole and sulphadimidine (in mixtures) and calamine and zinc oxide (in lotions) - a thickening agent is necessary to obtain a uniform product from which uniform doses can be removed.

Insolubility:

- It means the **inability of material to dissolve in a particular solvent system.**
- The liquid preparations containing **Indiffusible solids** such as chalk, aromatic chalk powder, acetyl salicylic acid, phenacetin, zinc oxide & calamine etc.
- **Suspending agent**
 - Increase the thickness of the preparation.
 - Uniform distribution of the insoluble substances which facilitating uniform measurement of each dose.
- E.g Acacia, Tragacanth, etc

Rx

Phenacetin	3 gm
Caffeine	1 gm
Orange syrup	12 ml
Water up to	90 ml

Directions for Pharmacist:

Make a mixture. Send 25 ml

Comments:

Physical incompatibility.

- In this prescription phenacetin is an indiffusible sub.
- Compound powder of **tragacanth** or **mucilage of tragacanth** is used as a suspending agent to make a stable suspension.

Menthol	5 gm
Camphor	5 gm
Ammonium chloride	30 gm
Light magnesium carbonate	60 gm

Directions for Pharmacist:

Make an Insufflations. Send 40 gm.

Comments:

Correct prescription.

Precipitation

Precipitation can result when the solvent in which the solute is insoluble is added to the solution.

Resins are normally not soluble in water. So, the tinctures of resins may form precipitate on addition to water.

- **Methods of Rectifying Insolubility:**
- **Co solvency :** *for e.g. we may use alcohol, propylene glycol, syrups.*
- **Complexation:** *for e.g. formation of tri-iodide complex, complexation of caffeine with Sodium Benzoate.*
- **Hydro trophy:** *for e.g. Hyoscamine with tween .*
 - Hydrotrophy is a solubilization process where by addition of second solutes results in an increase in the aqueous solubility of another solute.
- **Solubilization:** *for e.g. Fats soluble vitamins, certain antibiotics.*

2. Liquification of solids mixed in a dry state (eutexia)

Def.: it means that when two solid substances are mixed together, conversion to a liquid state take place.

It happens through the following methods:

1. Formation of liquid mixture: when the solid substance is soluble in another solid substance which lead to decrease of its melting point and conversion to a liquid in certain ratios.

When the substances with low melting points such as camphor, menthol and thymol are mixed together, a liquid mixture i.e. eutectic mixture is formed and this process is known as liquefaction.

2. Exit of crystalline water: By mixing hydrated crystals and dry crystals, crystalline water diffuse to dry crystals.

Examples:

Camphor ,menthol, phenol, thymol ,chloral hydrate, sodium salicylate, Aspirin , phenazone.

Methods Of Rectifying Liquefaction:

By the use of absorbent like kaolin, light magnesium carbonate.

Rx

Thymol	250 mg
Camphor	2 mg
Menthol	2 mg

Make powder.

- *Comments:* If these ingredients are triturated together, they will form an eutectic mixture.

Method-I:

All the ingredients are triturated.

An eutectic mixture (liquid) will be formed. The liquid is triturated with enough absorbent powder e.g. light kaolin or light magnesium carbonate, to give a free flowing powder

Method-II:

Each ingredient is triturated separately with small amount of adsorbent or diluent and then these powders are lightly mixed by tumbling action) and packed.

The diluent largely prevents contact between the ingredients and adsorbs any liquid that may be produced.

Rx

Chloral hydrate 250 mg

Prepare capsules. Supply 10 capsules.

Label: Take the capsules at night time.

Comment: Chloral hydrate is hygroscopic in nature. It will absorb moisture and soften the hard gelatin capsule shells and the shape of the capsule may change physically.

Remedy: An equal quantity of light magnesium oxide should be mixed with chloral hydrate.

Other adsorbents those may be used are kaolin, talc, starch etc.

Immiscibility of two or more liquids

- This manifestation appears clearly in emulsion, creams, lotions, some types of ointments.
- Separation in two phases is noticed in these pharmaceutical dosage forms.

The following factors lead to immiscibility:

1. Incomplete mixing
2. Addition of surfactant with:
 - Unsuitable concentration
 - False time of addition
 - Unsuitable for the type of emulsion
3. Presence of microorganisms
 - Some bacteria grow on constituents of mixture i.e. gelatin, Arabic gum
 - Others produce enzymes which oxidize the surfactant
4. Temperature

Storage must be in room temperature to prevent separation

Emulsification:

- Water and oil are immiscible in each other and they can be made miscible by the addition of Emulsifying agents. This is known as Emulsification.

- Oils are immiscible with water and hence combination of oily drugs with water produces a product possessing two separate layers.

Remedy: This problem can be overcome by emulsification or solubilization.

- Care must be taken when concentrated hydroalcoholic solutions of volatile oils such as *spirits and concentrated waters*, are used as adjuncts (e.g. as flavouring agents) in aqueous preparations. Large globules of oils may be separated.

Remedy: To prevent the formation of large globules, the hydroalcoholic solution should either be gradually diluted with the vehicle before admixture with the remaining ingredients or poured into the vehicle with constant stirring.

Methods for correction / prevention of physical incompatibility

A) Modified order of mixing:

- This method will vary often to overcome certain type of physical incompatibility. So this should be considered first.
- Example if salt is not soluble in alcohol, prescribed in hydro alcoholic liquids. So we dissolve salt first in water and then this solution should be mixed with the liquid prescribed.

Example:

- alcohol soluble substances to be dispensed in hydro alcoholic liquids. First dissolve in small amount of alcohol, then mixed with the prescribed solution. A general rule is to dissolve the prescribed substance first in the solvent in which it is most soluble and then add to this solution slowly with constant stirring so as to effect the gradual dilution in the liquid in which it is least soluble.

B) Alternate of solvents:

Solvents can be altered but to a limited extent.

Example:

- an alcohol soluble drug may be prescribed in elixir that does not have sufficient concentration of the alcohol to hold it in solution. It is then allowed to increase alcoholic concentration within a reasonable limit. So long as the volume of the prescription is not changed.
- Alcohol: it is a good solvent for many organic compounds such as alkaloid, camphor, phenol, tannins, balsams, resins and some other organic acids.
- Glycerin: it has solvent properties between those of alcohol and water.
- It is particularly good for dissolving tannins, boric acid, phenols, codeine and certain metallic salts.

C) Changing the form of ingredients:

A change in the form of the ingredients is often unable to obtain a better preparation. However, it should be clearly understood that this action is applicable only when the therapeutic action of ingredients not altered.

Example of prescription:

Phenobarbital gr vii (7 gran)

Sodium salicylate z I (1dram)

Aqua mentha preparata q.s to make z ii (2 ounce)

Phenobarbital is insoluble in aqueous solution. Sodium salt of Phenobarbital, which has the same action and the same dose, may be used in its place to obtain a clear solution.

Therapeutic Incompatibility

It may be the result of prescribing certain drugs to the patient with the intention to produce a specific degree of action but the nature or the intensity of the action produced is different from that intended by the prescriber.

CAUSES:

It may be due to the administration of :

- *Overdose or improper dose of a single drug.*
- *Improper Dosage form.*
- *Contraindicated drug.*
- *Synergistic and antagonistic drugs.*

(A) Over dose

-Additive and synergistic combinations:

There are certain drugs possessing similar pharmacological activity. If these drugs are combined together, they may produce additive or synergistic action. In such case advice of the physician is necessary.

e.g. Rx

Amphetamine sulphate	20 mg
Ephedrine sulphate	50 mg
Syrup q.s.	100 ml

Let a mixture be made

Label: Take 25 ml every four hours.

-Excessive daily dose

In this case the daily dose of drug is exceeded .

e.g. 1 **R_x**

Codeine phosphate	15 mg
-------------------	-------

Ammonium chloride 500 mg

Prepare capsules and supply 24 capsules.

Label: Two capsules to be taken every hour for cough.

Comment: The U.S.P. recommends that the prescribed dose should be taken after every four hours and not every hour. Hence the physician should be consulted.

Rx

Amphetamine Sulphate		20 mg
Ephedrine Sulphate		100 mg
Simple syrup	up to	100 ml

Directions for Pharmacist:

Make a mixture.

Comments: *Comment:* Both of the drugs are sympathetic stimulants and they are prescribed in their full dose. The formulation will produce additive overdose effect. Hence, The dose of individual drug should be reduced.

Therapeutic incompatibility.

(B) Under dose In this type of incompatibility, effect of one drug is lessen or antagonised by the presence of another drug.

This can be exemplified by combination of following types of drugs:

1. Stimulants like nux-vomica, strychnine sulphate, caffeine etc. with sedatives like barbiturates, paraldehyde etc.
2. Sympathomimetic or adrenergic like ephedrine, nor-adrenaline with sympatholytic drugs like ergotamine.
3. Sympathetic stimulants like methamphetamine with parasympathetic stimulants like pilocarpine.
4. Purgatives like castor oil, liquid paraffin etc with antidiarrheal agents like bismuth carbonates.
5. Acidifiers like dilute hydrochloric acid and alkalisers like sodium bicarbonate, magnesium carbonate.

Rx

Aspirin 300 mg

Probenecid 500 mg

Prepare capsules.

Label: One capsule a day for gout.

Aspirin is an NSAID given to reduce the pain and swelling in case of gout attack.

Probenecid blocks the active reabsorption of uric acid from the lumen of nephron, but salicylates (aspirin) blocks this action of probenecid.

Hence, both of the drugs are antagonistic to each other, so its combination is therapeutically useless.

(C) Improper consumption by the patient:

In certain prescription some special directions should be written. If the patients are not advised the drugs may not produce the desired action due to low bioavailability.

e.g. **Rx**

Tetracycline hydrochloride 250 mg

Prepare capsules. Supply 10 capsules.

***Label:* Take one capsule every six hourly with milk.**

***Comments:* Calcium present in milk inactivates the tetracycline, hence a patient may not get any therapeutic effect if he/she takes the capsule with milk.**

***Remedy:* The pharmacist should advise the patient to take the capsule with water and not with milk. The patient should not take antacid containing calcium salts.**

(D) Contra-indicated drugs

Certain drugs should not be given in particular disease condition

e.g.

- (i) corticosteroids are contraindicated in patients with peptic ulcer.
- (ii) Vasoconstrictors are contraindicated in hypertensive patients
- (iii) Some drugs should not be given in asthmatic patients e.g. barbiturates, morphine etc.
- (iv) If a person is allergic to a drug (e.g. penicillin injection) then it should not be given to the patient.

Certain combination of drugs are contraindicate:

Rx

Sulphadiazine	0.25 g
Sulphamerazine	0.25 g
Ammonium chloride	0.50 g

Prepare capsules

Label: Take two capsules six hourly for cough.

Comment: In this prescription ammonium chloride is a urinary acidifier and it could cause deposition of sulphonamide crystals in the kidney.

THANK YOU